國立臺灣大學管理學院資訊管理研究所

碩士論文

Department of Information Management

College of Management

National Taiwan University

Master’s Thesis

探索端對端生成式語音檢索

Exploring End-to-End Speech Generative Retrieval

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中華民國 114 年 9 月

September, 2025

doi:10.6342/NTU202504440

Acknowledgements

感謝碩士過程中家人的支持和教授的指導，也感謝郭子麟給的題目靈感。然後模型名其實是 Chaewon: SpeeCH-AwarE Generative Retrieval for open-domain Question ansWering Over discrete uNits，紀念偶像給我的精神支持。

doi:10.6342/NTU202504440

doi:10.6342/NTU202504440

摘要

本文提出 SpeechGR，一個針對口語查詢與段落設計的端到端生成式檢索系統。與傳統依賴語音辨識或密集向量索引的傳統檢索方法不同，SpeechGR 直接處理原始語音，將查詢與文件表示為離散語音標記。隨後，一個生成式語言模型學習從這些標記中產生文件識別碼，無需額外的文字語音監督或外部索引。在 SLUE-SQA-5 數據集上進行評估，SpeechGR 與多種基準模型進行比較，包括 BM25、密集檢索模型以及專為語音設計的 SpeechDPR。實驗結果顯示，SpeechGR 的檢索準確度與 SpeechDPR 相當，其 Top-1 與 Top-20 準確度差距在一百分點以內，同時保持完全語音辨識且無索引的設計。消融實驗進一步揭示了離散聲學標記提取層、識別碼設計以及預訓練時長對性能的影響。通過展示語音專用生成式檢索的可行性，SpeechGR 提供了一種新穎的資訊檢索方法，繞過了傳統瓶頸，如自動語音辨識依賴和索引管理。本研究推進了生成式檢索與無文字自然語言處理領域，為未來多模態、無監督檢索系統的探索奠定了基礎。

關鍵字：語音、生成式檢索

doi:10.6342/NTU202504440

doi:10.6342/NTU202504440

Abstract

We presents **SpeechGR**, an end-to-end generative retrieval system tailored for spoken queries and passages. Unlike conventional retrieval methods that depend on text transcripts or dense vector indices, SpeechGR processes raw speech directly, representing queries and documents as discrete acoustic tokens. A generative language model then learns to produce document identifiers from these tokens, eliminating the need for text supervision or external indices.

Evaluated on the SLUE-SQA-5 dataset, SpeechGR is benchmarked against diverse baselines, including BM25, dense retrieval models, and the speech-specific SpeechDPR. Experimental results show that SpeechGR achieves retrieval accuracy comparable to SpeechDPR, with Top-1 and Top-20 accuracies within one percentage point, while maintaining a fully transcript-free and index-free design. Ablation studies further highlight the influence of discrete unit extraction layers, identifier design, and pre-training duration on performance.

By demonstrating the viability of speech-only generative retrieval, SpeechGR offers a

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novel approach to information retrieval that bypasses traditional bottlenecks like ASR dependency and index management. This research advances the fields of generative retrieval and textless NLP, providing a foundation for future exploration of multi-modal, supervision-free retrieval systems.

Keywords: Speech, GR, Generative Retrieval

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Chapter 1 Introduction

In recent years, information retrieval (IR) has undergone significant advancements, driven by innovations in machine learning and natural language processing (Zhu et al., 2023). Traditional retrieval systems, such as BM25 (Robertson and Zaragoza, 2009), rely on lexical matching between queries and documents, which limits their ability to handle semantic variations or non-textual inputs like speech. Dense retrieval methods have mitigated some of these limitations by embedding queries and documents into continuous vector spaces, capturing deeper semantic relationships (Chen et al., 2024; Karpukhin et al., 2020). However, these approaches often require substantial computational resources and external indices, posing challenges for scalability and adaptability to diverse modalities. Moreover, heterogeneous ranking components in dense retrieval, as well as traditional retrieval pipelines involving indexing, retrieving, and reranking, are usually difficult to optimize in an end-to-end manner towards a global objective.

A critical limitation in current retrieval systems is their dependence on text, which is not universally available across the world's languages. Of the over 7,000 languages spoken globally, nearly half lack standardized written forms or sufficient textual data for training robust NLP or retrieval models (Ethnologue, 2025). This absence of text resources severely restricts the applicability of text-based retrieval systems in diverse linguistic contexts, particularly for low-resource or unwritten languages. Consequently, there is a need

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for retrieval systems that can operate directly on spoken inputs, bypassing the requirement for text transcripts entirely.

Generative retrieval (GR) has emerged as a promising alternative, leveraging generative models to directly produce document identifiers from queries (Kuo et al., 2024; Li et al., 2024a) . This index-free approach simplifies retrieval systems and holds significant potential for handling multi-modal data, such as speech, where text supervision may be unavailable or impractical. Despite its potential, generative retrieval for spoken content remains underexplored, particularly in end-to-end settings that eliminate text intermediaries.

We introduces **SpeechGR**, a novel end-to-end generative retrieval system designed for spoken queries and passages. SpeechGR represents both queries and documents as discrete acoustic tokens derived from raw audio, using a generative language model to map spoken inputs directly to document identifiers. Unlike existing methods that rely on automatic speech recognition or dense embeddings, SpeechGR operates without text supervision or external indices, encapsulating all retrieval logic within the model's parameters. This text-free approach is particularly advantageous for addressing the challenges posed by languages without written forms, enabling direct processing of spoken content and broadening the accessibility of information retrieval systems.

The primary objectives of this research are:

- To design and implement a generative retrieval model that processes speech data directly, eliminating the need for text transcripts.
- To assess SpeechGR's performance against established baselines.
- To investigate the effects of key design choices—such as discrete unit extraction,

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identifier design, and training strategies—on retrieval accuracy and efficiency for GR on speech.

This study evaluates SpeechGR on the SLUE-SQA-5 dataset (Shon et al., 2022), demonstrating competitive performance against state-of-the-art methods while maintaining a fully textless, index-free design. The contributions of this work include: (1) A novel framework for end-to-end speech-to-speech generative retrieval, eliminating dependency on automatic speech recognition (ASR) and external indices. (2) Advancements in speech-based generative retrieval, enabling direct processing of raw audio for languages with limited or no textual resources. (3) Insights into optimizing generative retrieval for speech inputs, particularly through the analysis of discrete unit extraction, identifier design, and pre-training strategies.

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Chapter 2 Literature Review

2.1 From Sparse to Dense Retrieval

Background Information retrieval (IR) systems aim to fulfill a user's information need by processing a query—expressed as text, speech, images, or other modalities—and returning a ranked list of relevant items from a large corpus. The effectiveness of these systems is typically evaluated using metrics such as precision@k, recall@k, or Hit@k, which assess the presence of relevant items within the top k results. Retrieval methods primarily differ in their approaches to representing queries and corpus items and in the techniques used to score their relevance. Over time, the field has progressed from sparse retrieval, which relies on lexical matching, to dense retrieval, which leverages semantic embeddings, and most recently to index-free generative retrieval, which eliminates external indices. The following sections trace this evolution, highlighting key developments and their implications for speech-based retrieval.

Sparse Retrieval. Sparse retrieval ranks documents solely by their lexical overlap with the query, representing both sides as bags of words. The most widely used scoring function in this family is BM25 (Robertson and Zaragoza, 2009). For a query q and a document d,

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$$\text{BM25}(q,d) = \sum_{t \in q} \text{IDF}(t) \, \frac{f(t,d)\,(k_1+1)}{f(t,d) + k_1 \left[1 - b + b \frac{|d|}{\text{avgdl}}\right]},$$

where $f(t,d)$ is the term frequency of t in d, $|d|$ is the document length, avgdl is the average length of all documents, k_1 controls term-frequency saturation, and b sets the strength of length normalization. BM25 relies entirely on word matching. If a passage expresses the same idea with synonyms, paraphrases, or in another language, its score becomes zero. The approach also extends poorly to non-text modalities—speech, images, or multimodal inputs—because there is no obvious notion of "term frequency." The limitations of sparse retrieval have prompted a shift toward dense retrieval. In this approach, queries and documents are represented as continuous vectors in a shared embedding space, where similarity measures capture semantic relationships rather than surface-level term overlaps.

Dense Retrieval. Dense retrieval addresses the limitations of lexical matching by representing queries and documents as continuous vectors in a shared embedding space, where semantic similarity drives ranking. This approach employs encoder architectures—either a single shared encoder or separate encoders for queries and documents—to generate dense embeddings, enabling efficient similarity-based retrieval. The rise of large language models (LLMs), such as BERT (Devlin et al., 2019), RoBERTa (Liu et al., 2019), and their multilingual variants, has significantly advanced this paradigm by providing contextual representations that capture rich semantic information, improving performance across diverse queries and languages.

A prominent example is DPR Karpukhin et al. (2020), which fine-tunes two BERT-based encoders on question-answer pairs for open-domain question answering. DPR optimizes

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the negative log likelihood of the positive query-passage pair using the following loss function:

$$L(q_i, p_i^+, p_{i,1}^-, ..., p_{i,n}^-) = -\log \frac{e^{sim(q_i,p_i^+)}}{e^{sim(q_i,p_i^+)} + \sum_{j=1}^{n} e^{sim(q_i,p_{i,j}^-)}}$$

For each training instance $D = \{q_i, p_i^+, p_{i,1}^-, ..., p_{i,n}^-\}$, contains a query q_i, its corresponding passage p_i^+, and negative samples $p_{i,1}^-, ..., p_{i,n}^-$, DPR employs in-batch negative training by using passages paired with other queries in the same mini-batch as negative samples, optimizing the model to maximize similarity between the query and its positive passage while minimizing similarity with these negative passage. At inference time, each passage embedding **d** is pre-computed and stored in an ANN index such as FAISS (Douze et al., 2024). A query embedding **q** is generated on the fly, and the top-k nearest neighbours are returned as the retrieval result. DPR significantly outperforms BM25 in passage retrieval for open-domain question answering, demonstrating the power of dense retrieval.

Current state-of-the-art text-retrieval models, such as BGE-M3 Chen et al. (2024), leverage large-scale pre-trained encoders to support both sparse and dense retrieval. BGE-M3 produces versatile embeddings that achieve top performance on benchmarks like BEIR (Thakur et al., 2021) and MIRACL Zhang et al. (2023) without requiring task-specific fine-tuning. Its ability to handle multilingual and multi-granularity retrieval highlights the flexibility of modern dense retrieval systems.

Despite their accuracy, dense retrieval methods face significant challenges. Storing high-dimensional embeddings for large corpora demands substantial memory, often requiring hundreds of gigabytes. Building and updating ANN indices is computationally intensive, and performance relies on continual hard-negative mining and retraining as corpora

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evolve. Moreover, the heterogeneous nature of dense retrieval systems, where index retrieval and ranking components pursue distinct objectives, complicates end-to-end optimization for both ranking performance and storage efficiency (Tang et al., 2024).

2.2 Generative Retrieval

2.2.1 Core Concept and Pipeline

Generative retrieval (GR) redefines information retrieval as a sequence generation task, moving away from similarity-based matching used in sparse and dense retrieval methods. Instead of relying on external indices, GR employs a single encoder-decoder model g_θ, parameterized by θ, to internalize a corpus within its weights and generate unique identifiers—short token sequences, often termed DocIDs—that represent target passages (Kuo et al., 2024; Li et al., 2024a; Tang et al., 2024). This approach eliminates the need for external storage and enables end-to-end optimization, addressing key limitations of dense retrieval systems.

Consider a corpus $\mathcal{C} = \{(p_i, y_i)\}_{i=1}^{N}$, where p_i is a passage and y_i is its unique identifier, and a retrieval dataset $\mathcal{Q} = \{(q_j, p_j)\}_{j=1}^{N}$, where q_j is a query associated with passage p_j. The GR model retrieves p_j by generating its identifier y_j based on q_j. Training typically involves two tasks:

1. **Indexing:** feed the passage tokens into the encoder and train the decoder to memorize its identifier: $g_\theta(p_i) \rightarrow y_i$.

2. **Retrieval:** feed a query q_j and decode the most likely identifier(s): $g_\theta(q) \rightarrow y_j$.

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Tay et al. (2022) introduced DSI, a pioneering GR framework using a T5 encoder-decoder model. DSI optimizes two auto-regressive cross-entropy losses:

$$\mathcal{L}_{\text{index}} = -\sum_{(p_i,y_i)\in\mathcal{C}}\sum_{\ell=1}^{|y_i|}\log P_\theta\big(y_{i,\ell} \mid y_{i,<\ell}, p_i\big),$$

$$\mathcal{L}_{\text{retr}} = -\sum_{(q_j,y_j)\in\mathcal{Q}}\sum_{\ell=1}^{|y_j|}\log P_\theta\big(y_{j,\ell} \mid y_{j,<\ell}, q_j\big),$$

enabling a single parameter set to serve as both index and retriever.DSI advances generative retrieval by exploring innovative identifier designs, discussed in detail in subsequent sections. DSI demonstrates superior performance compared to SentenceT5 and sparse retrieval on the Natural Questions (NQ) dataset, highlighting GR's potential.

At inference, GR methods typically use constrained beam search with a prefix tree of valid identifiers to generate the identifier autoregressively (Cao et al., 2020). The i-th token of the identifier is produced as:

$$\mathbf{h}_i^d = Decoder(c_{<i}^d; Encoder(q))) \in \mathbb{R}^D.$$

where $c_{<i}^d$ is feed into the decoder with encoder query vector to calculate cross-attention.

DSI demonstrates several core concept and advantage of the GR systems:

- **Index-free:** No external ANN or inverted index is required post-training, reducing storage and maintenance costs.
- **Unified memory:** A single model encapsulates both passage and query knowledge, simplifying the retrieval pipeline.

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- **End-to-End Optimization:** A unified optimization goal enables seamless training, overcoming the challenges of heterogeneous components in dense retrieval.

These properties make GR particularly promising for applications like speech retrieval, where text dependency limits traditional methods. The following sections discuss identifier design, data augmentation, coverage, optimization, and training paradigms in detail.

2.2.2 Identifier Design

In generative retrieval (GR), the design of document identifiers—sequences generated to represent passages—plays a critical role in balancing uniqueness, semantic alignment with queries, and decoding efficiency. Identifiers must uniquely map to passages while enabling the model to learn and generate them effectively from input queries or passages. This section reviews three identifier categories: number-based, word-based, and multiple identifiers, highlighting their strengths and limitations.

2.2.2.1 Number-based Identifiers

Unstructured atomic integers. The original DSI framework assigns each passage a unique integer, treating the decoder's final hidden state as a logit over a vocabulary of all document identifiers (Tay et al., 2022). This approach simplifies identifier generation but requires a Softmax operation over the entire corpus size $|\mathcal{C}|$, limiting scalability to corpora with fewer than a few million documents due to computational constraints.

Naïvely tokenized integers. To address scalability, DSI also explores tokenizing integer identifiers into decimal digits (e.g., 243168 → 24 316 8) (Tay et al., 2022; Zhuang

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et al., 2022). This reduces the output vocabulary size and eliminates the need for a unique vector per document. However, these identifiers lack semantic structure, offering little information for the model to learn meaningful relationships between passages.

Semantically structured numeric paths. Tay et al. (2022) further proposes hierarchical clustering of dense passage embeddings to create numeric identifiers with semantic meaning. Given a corpus, documents are recursively clustered into groups (e.g., 10 clusters), with each cluster assigned a numeric prefix. This process continues for clusters exceeding a size threshold c, producing identifiers that reflect semantic proximity and reduce the search space during decoding. While effective, this method requires additional preprocessing to generate embeddings.

Product-quantization codes. Zhou et al. (2022) compresses dense passage embeddings using product quantization (PQ), yielding short, fixed-length code strings as identifiers. These codes preserve semantic proximity, enabling efficient retrieval with compact representations. However, generating PQ codes demands significant computational resources for embedding compression.

Number-based identifiers are straightforward to generate and ensure uniqueness but often lack semantic richness, limiting interpretability and scalability for large corpora.

2.2.2.2 Word-based Identifiers

Titles and entity names. Cao et al. (2020) uses Wikipedia titles as identifiers, producing human-readable outputs that align closely with natural-language queries. This approach enhances interpretability but risks title duplication and is restricted to datasets with reliable

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metadata, such as Wikipedia, limiting its applicability to other corpora.

URLs. For web-based corpora, tokenized URLs serve as unique identifiers with moderate semantic content. Ren et al. (2023) solely used tokenized URLs as identifier. Zhou et al. (2022) combines URL and title to balance uniqueness and meaning. These methods perform well on datasets like MS MARCO but depend on stable, metadata-rich sources, which may not be available in all domains.

Pseudo-queries Identifier. Tang et al. (2023b) employs Doc2Query (Cheriton, 2019) to generate one natural-language query for every passage and then uses that query text itself as the document identifier. Because Doc2Query is stochastic, different passages sometimes receive identical pseudo-queries; when an identifier collides, the system resolves the tie by returning all passages that share the same string in random order. Although this strategy embeds rich semantics in the id and simplifies decoding to ordinary language generation, it sacrifices the one-to-one mapping between id and passage, making evaluation and downstream usage more complex.

Important term sets. Zhang et al. (2024) chooses the top-N salient terms for each document and regards any permutation of those terms as a valid identifier. Allowing multiple orderings sharply lowers false-pruning errors during decoding and yields noticeably higher retrieval accuracy on the NQ-320K benchmark . The method, however, depends on a pre-trained relevance scorer to rank candidate terms.

Word-based identifiers offer semantic richness and human-readability but depend on text-based metadata or additional processing (e.g., query generation, term ranking), restricting

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their use to text-rich datasets.

2.2.2.3 Multiple Identifiers

Recent GR systems explore assigning multiple identifiers per passage to capture diverse representations. Bevilacqua et al. (2022b) selecting those with high query overlap during training. Chen et al. (2023) use BERT to determine important n-grams as identifier. Li et al. (2023) combine title, sub-strings, and pseudo-queries as identifier. These approaches enhance retrieval flexibility by representing passages from multiple perspectives but increase memory and computational demands compared to single-identifier systems. Additionally, managing multiple identifiers introduces challenges in training stability and decoding ambiguity.

Identifier design in GR involves a trade-off among uniqueness, semantic alignment, and decoding efficiency. Number-based identifiers ensure uniqueness but lack interpretability and scalability. Word-based identifiers embed rich semantics but rely on text metadata, limiting their applicability to languages or datasets without written forms. Multiple identifiers offer comprehensive representations but increase resource demands. For our system, SpeechGR, we adopt short string identifiers encoding article and sentence offsets, leveraging the corpus' s inherent structure to ensure uniqueness without text dependency or extensive preprocessing. This choice suits speech-based retrieval by avoiding reliance on text metadata, aligning with the goal of processing raw audio for languages lacking written forms. Further discussion of identifier choices follows in subsequent sections.

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2.2.3 Data augmentation

A key challenge in generative retrieval (GR) is the data distribution mismatch between indexing and retrieval tasks. Indexing typically involves long, detailed passages, while retrieval relies on short, concise queries, impairing the model' s ability to map queries to passage identifiers effectively (Zhuang et al., 2022). Data augmentation, particularly query generation (QG), addresses this gap by aligning the data distributions of indexing and retrieval tasks, enhancing GR performance.

Query Generation Query generation (QG) augments GR training by generating pseudo-queries that mimic real user queries, bridging the distributional gap between passages and queries. For example, DSI-QG Zhuang et al. (2022) and NCI (Wang et al., 2022) use a query-generation model, such as DocT5Query (Cheriton, 2019), to produce multiple pseudo-queries for each passage. These pseudo-queries replace original passage-identifier pairs $\{p, y\}$ with pseudo-query-identifier pairs $\{\tilde{q}, y\}$ during indexing, aligning the indexing task with the retrieval setting. This approach improves retrieval hit rate by training the model on inputs that resemble actual queries, as demonstrated by enhanced performance on benchmarks like Natural Questions compared to non-augmented GR models.

While QG is effective for text-based GR, it relies on text metadata or query-generation models, limiting its applicability to speech-based retrieval or datasets lacking textual queries, such as those for unwritten languages. In our system, SpeechGR, we bypass QG by directly processing raw audio to generate identifiers, avoiding text dependency and aligning with the goal of textless retrieval.

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2.2.4 Training Paradigms and Optimization Strategies

Training generative retrieval (GR) systems requires optimizing models to generate accurate document identifiers while ensuring effective ranking of relevant passages. Early GR approaches rely on token-level cross-entropy objectives to train encoder-decoder models to generate valid identifier sequences, but these provide no direct signal for ranking relevant documents over irrelevant ones (Bevilacqua et al., 2022a; Tay et al., 2022; Wang et al., 2022; Zhuang et al., 2022). Recent advancements incorporate pairwise ranking losses to improve retrieval performance, a strategy we explore in our ablation study for SpeechGR to optimize identifier generation from audio inputs (Li et al., 2024a; Tang et al., 2024).

Li et al. (2024c) introduces a pairwise ranking loss to enhance ranking performance while retaining the standard cross-entropy objective. After pre-training with cross-entropy, them fine-tunes the model in a second phase, maximizing a fixed margin between the relevance scores of a positive passage p^+ and a mined negative passage p^-:

$$\mathcal{L}_{rank} = \max(0, s(q, p^-) - s(q, p^+) + m), \quad \mathcal{L} = \mathcal{L}_{CE} + \mathcal{L}_{rank},$$

where $s(q, p)$ is the relevance score for query q and passage p, and m is a fixed margin. This fine-tuning improves ranking accuracy without increasing inference time, achieving higher retrieval performance on benchmarks like Natural Questions compared to cross-entropy alone.

Zeng et al. (2023) propose RIPOR, extending pairwise ranking to the prefix level to address beam search pruning issues, where relevant documents may be lost if their identifier prefixes are pruned early. At each decoding position i, RIPOR enforces a margin between the prefix scores of a positive identifier c_{d^+} and a negative identifier c_{d^-}:

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$$\mathcal{L}^i_{rank}(q, c_{d^+}, c_{d^-}) = (S^i_{prefix}(q, c_{d^+}) - S^i_{prefix}(q, c_{d^-}) - \alpha_i T_{(q,d^+d^-)})^2$$

, where S^i_{prefix} is the relevance score for the query and prefix at position i, $T_{(q,d^+,d^-)}$ is a target margin predicted by a teacher cross-encoder, and α_i emphasizes early prefixes to prevent pruning errors. This prefix-level pairwise ranking enables RIPOR to outperform dense retrievers without requiring external indices, as demonstrated on datasets like MS MARCO.

Pairwise ranking losses, as used in LTRGR and RIPOR, are critical for optimizing GR systems to prioritize relevant documents, a strategy we adopt in SpeechGR’ s ablation study to enhance audio-based identifier generation. Unlike text-based GR, SpeechGR applies these losses to discrete acoustic tokens, bypassing text dependency and enabling retrieval for languages without written forms.

2.3 Non-Text Generative Retrieval

Generative retrieval offers a flexible framework for processing diverse modalities, as its sequence generation approach can, in principle, produce identifiers for any data type, from text titles to audio or image tokens. In practical applications, users often need to retrieve non-text objects—such as images, audio clips, or videos—prompting research into GR systems that handle non-text corpora. Unlike traditional retrieval, which requires separate indices for text, vision, or audio, GR enables a unified model to internalize and retrieve across these modalities.

Current GR research for non-text modalities primarily uses textual queries to generate identifiers for non-text objects. Li et al. (2024b) demonstrates that a multimodal large

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language model (LLM) can memorize an image corpus within its parameters and autoregressively predict image identifiers from text prompts, showcasing cross-modal GR capabilities. Long et al. (2024) maps visual object features into an LLM's textual embedding space via prefix-tuning, allowing the model to fuse image cues with text queries and generate knowledge clues for retrieving relevant documents without additional neural computation. Similarly, Fang et al. (2024) employs K-means and RQ-VAE to derive coarse- and fine-grained tokens for indexing images, audio, and video, enabling text-to-non-text retrieval with high accuracy.

Despite these advancements, non-text GR remains text-centric, relying on textual queries and operating within language-based embedding spaces. This limits the ability to handle non-text queries, such as retrieving an audio clip from an image or a video from a spoken query, which remains an unexplored challenge. In our system, SpeechGR, we address this gap by enabling speech-to-speech retrieval, generating identifiers directly from raw audio queries and passages without text intermediaries. By adapting pairwise ranking losses, as explored in our ablation study, SpeechGR optimizes identifier generation for acoustic tokens, supporting retrieval for languages lacking written forms. These textless capabilities distinguish SpeechGR from text-centric non-text GR approaches, paving the way for truly modality-agnostic retrieval systems.

2.4 Speech Retrieval

Speech retrieval extends text-based generative retrieval (GR) to process raw audio as queries, passages, or both, addressing the challenge of retrieving content from spoken data without relying on text transcripts. This emerging field aims to develop models that

doi:10.6342/NTU202504440

directly handle acoustic inputs, particularly for languages lacking written forms, where automatic speech recognition (ASR) is impractical. Recent research has progressed from ASR-dependent systems to unsupervised and end-to-end approaches, reducing text dependency and enabling more robust speech retrieval systems.

Early work focused on mitigating ASR noise. AVATAR augments a dual-encoder retriever with an adversarial objective, training the speech encoder to map semantically equivalent utterances to a shared embedding space despite transcription errors (Wang et al., 2023). By pairing spoken queries with corrupted text versions, AVATAR improves retrieval accuracy without modifying the text encoder, maintaining compatibility with text-based systems.

Advancing toward textless retrieval, SpeechDPR pioneers audio retrieval without parallel speech–text supervision (Lin et al., 2024). It distills sentence-level semantics from an unsupervised ASR model (Baevski et al., 2021) and a text-only dense retriever, fine-tuning with a contrastive loss similar to DPR. This approach achieves competitive performance using only self-supervised acoustic signals and minimal textual guidance, highlighting the potential for supervision-free speech retrieval.

Recent end-to-end frameworks further reduce reliance on ASR. Min et al. (2024) retrieves audio passages directly from text queries using a speech language model, bypassing ASR to improve performance over traditional pipelines. Chen et al. (2025) integrates audio and text in a unified embedding space for multimodal retrieval, generating contextually grounded responses with enhanced efficiency. However, these methods primarily use text queries, limiting their applicability to fully speech-based scenarios.

Following advancements in speech retrieval, SpeechRAG Min et al. (2024) introduces an end-to-end framework that bypasses automatic speech recognition to directly retrieve

doi:10.6342/NTU202504440

audio passages from text queries and generate answers using a speech language model, enhancing performance over traditional ASR-based systems. WavRAG Chen et al. (2025) proposes a novel RAG framework for spoken dialogue systems, integrating audio and text modalities in a unified embedding space to retrieve multimodal knowledge and generate contextually grounded responses, achieving improved efficiency and accuracy compared to ASR-dependent pipelines.

Our system, SpeechGR, addresses this gap by enabling end-to-end speech-to-speech retrieval, generating identifiers directly from raw audio queries and passages without text intermediaries. Leveraging pairwise ranking losses, as explored in our ablation study, SpeechGR optimizes identifier generation for acoustic tokens, achieving robust performance on datasets like SLUE-SQA-5 and supporting retrieval for unwritten languages. This textless approach distinguishes SpeechGR from text-centric speech retrieval systems, advancing the field toward fully modality-agnostic GR.

2.5 Discrete Speech Units and Textless NLP

Textless natural language processing (NLP) processes raw speech directly, bypassing text transcriptions, which is critical for low-resource languages with limited or no written data. Discrete speech units—tokenized representations derived from clustering self-supervised audio features—enable language models (LMs) to handle end-to-end speech tasks by encoding acoustic information into sequences compatible with generative frameworks. These units are particularly promising for speech retrieval, where direct audio processing can eliminate the need for text intermediaries, supporting applications in unwritten languages.

doi:10.6342/NTU202504440

A foundational study, GSLM (Lakhotia et al., 2021), uses discrete speech units generated via self-supervised learning and k-means clustering to train a generative LM for speech task. GSLM excels in speech resynthesis and continuation, demonstrating that discrete units capture rich linguistic patterns without text supervision, paving the way for textless NLP applications. Lin et al. (2022) extends this approach to textless spoken question answering, processing audio queries and documents with a pre-trained LM to achieve robust performance in noisy or low-resource settings, surpassing ASR-based baselines. Similarly, Shon et al. (2024) integrates discrete units with LMs for spoken language understanding tasks, such as intent classification, showing versatility across diverse audio inputs

Discrete speech units are increasingly applied to speech tasks due to their compatibility with language models (LMs), leveraging the benefits of LM pre-training for fine-tuning on applications like speech resynthesis, question answering, and spoken language understanding. Their ability to encode acoustic features into discrete tokens makes them promising for emerging tasks, such as speech retrieval, where direct audio processing can support low-resource, unwritten languages, opening new research directions in textless NLP.

doi:10.6342/NTU202504440

Chapter 3 Methodology

We introduce SpeechGR, an end-to-end system for speech-to-speech retrieval that generates passage identifiers directly from spoken queries without text supervision or external indices. Unlike prior approaches reliant on automatic speech recognition (ASR) or dense embeddings, SpeechGR encodes queries and passages as discrete acoustic tokens and employs a generative language model to map queries to identifiers. This textless framework internalizes all retrieval logic within the model's parameters, enabling retrieval for low-resource, unwritten languages. The retrieval process, illustrated in Figure 3.1 , involves transforming audio inputs into discrete tokens, adapting a generative model for speech, and training it to associate queries with identifiers. The following sections detail the key components: discrete speech unit representation, identifier design, model architecture, training objectives, and inference strategy.

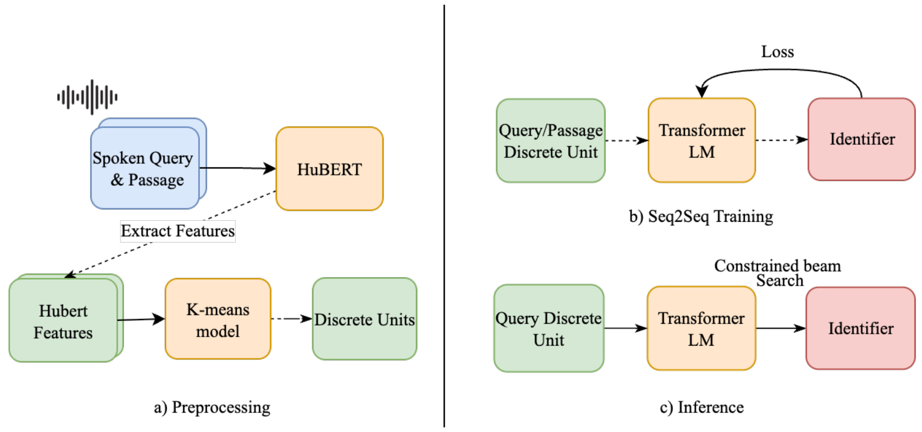

Figure 3.1: End-to-End Speech Generative Retrieval (SpeechGR) Pipeline

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3.1 Speech Input Representation

SpeechGR converts audio waveforms into discrete units for generative retrieval, processing each waveform into frame-level embeddings $\mathbf{z} = \{\mathbf{z}_1, \mathbf{z}_2, ..., \mathbf{z}_t\}$ with a pre-trained HuBERT model (Hsu et al., 2021). These embeddings $\mathbf{z}_t \in \mathbb{R}^d$ are quantized using a k-means codebook $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2, ..., \mathbf{c}_k\}$, where discrete units u_t are assigned as:

$$u_t = \arg\min_{j\in\{1,...,K\}} \|z_t - c_j\|_2^2,$$

with K as the codebook size, producing a token sequence $\mathbf{u} = (u_1, u_2, ..., u_t)$. To enhance sequence efficiency, we apply deduplication to collapse consecutive repeated units and use Byte Pair Encoding (BPE) to merge frequent adjacent units into higher level tokens (Wang et al., 2024). We also experiment with different HuBERT layers, where earlier layers focus on acoustic details and deeper layers capture semantic information, to optimize retrieval performance. Given the lack of speech-text pairs, we explore a Windowed Q-Former (Pan et al., 2023) to compress discrete unit sequences. This applies attention mechanisms over fixed-size windows, generating shorter representations for encoding, as a potential enhancement for compatibility with the generative model. We evaluate its impact on retrieval performance as part of our experimental analysis.

3.2 Identifier Design and Evaluation

SpeechGR requires unique identifiers to map spoken queries to specific passages. We design the system to use corpus-provided document IDs, short character strings encoding metadata (e.g., article prefix and sentence offset), which preserve corpus structure and sup-

doi:10.6342/NTU202504440

port future additions. For comparison, we test Unstructured Atomic Identifiers, assigning consecutive integers to passages—motivated by their effective use in visual retrieval—though they lack contextual information (Li et al., 2024b). Both approaches are evaluated to enable the sequence-to-sequence model to generate valid passage identifiers.

3.3 Model Architecture and Pre-training

SpeechGR employs the T5 family, including t5, mt5, and flan-t5 variants (Chung et al., 2024; Raffel et al., 2020; Xue et al., 2021), to support generative retrieval across different parameter scales. To incorporate discrete units from 3.1, we treat these units—discrete representations of raw audio extracted using HuBERT features and k-means clustering—as acoustic tokens, expanding the model's embedding lookup table by adding one embedding per acoustic token. Each new embedding is initialized by copying the vector of an unused text token from the original T5 vocabulary, enabling the model to process acoustic token sequences as input for generative retrieval.

For speech adaptation, we introduce a pre-training phase before retrieval fine-tuning, enabling the model to internalize the corpus index and adapt to discrete unit sequences. We use the span-corruption objective, where contiguous token spans are replaced with sentinel tokens, and the model reconstructs them in order, following the original T5 approach. The pre-trained model is then fine-tuned end-to-end on spoken query-passage pairs using the identifier-generation loss detailed in 3.4.

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3.4 Training Objective

SpeechGR is trained to simultaneously index passages and retrieve matching identifiers from spoken queries. The core objective uses a sequence-to-sequence cross-entropy (CE) loss, minimizing:

$$\mathcal{L}_{CE} = -\sum_{\ell=1}^{L} \log P(y_\ell^+ \mid y_{<\ell}^+, \mathbf{q}),$$

where $\mathbf{q}$ is the spoken query, $\mathbf{y}^+ = (y_1^+, ..., y_L^+)$ is the gold identifier, L is the identifier length, and $P(\cdot)$ denotes the softmax probability. As part of the ablation study, we also explore a ranking loss to assess its impact on retrieval performance.

3.5 Inference Strategy

At inference, SpeechGR converts each spoken query into a discrete-unit sequence and processes it through the encoder. The decoder generates the passage identifier autoregressively using constrained beam search, guided by a trie of valid corpus identifiers. At each step, logits for invalid token paths are masked, and beam pruning retains the B most-probable prefixes, which serve as the system's top-k retrieval results.

3.6 Summary

This chapter has outlined the methodology for SpeechGR, the first end-to-end speech generative retrieval system. By converting spoken queries into discrete units, designing unique identifiers, adapting the T5 architecture with speech-aware pre-training, optimizing with a cross-entropy loss, and employing a trie-guided inference process, SpeechGR

doi:10.6342/NTU202504440

internalizes all retrieval logic without external indices. These methods collectively address the challenge of low-resource, unwritten languages, setting the stage for experimental validation in Chapter 4, where the system's performance, including ablation studies, will be assessed.

doi:10.6342/NTU202504440

doi:10.6342/NTU202504440

Chapter 4 Experiments & Results

4.1 Data

SpeechGR adopts the speech-to-speech retrieval setting from SpeechDPR (Lin et al., 2024), using spoken queries from SLUE-SQA-5 (Shon et al., 2022) and passages from Spoken Wikipedia recordings linked to the dataset. Originally designed for spoken question answering, SLUE-SQA-5 queries are paired with their corresponding passages as retrieval targets. To ensure a one-to-one identifier mapping, only Spoken Wikipedia segments present in SLUE-SQA-5 are indexed. Speech are converted into discrete-unit sequences, deduplicated, and processed with BPE, with no textual transcripts or speech-text pairs used during training or evaluation.

4.2 Baseline

SpeechGR is evaluated against a select set of models representing key retrieval paradigms, as detailed in Table 4.1. These include SpeechDPR, a state-of-the-art speech-only end-to-end retriever, and BM25, a classical sparse text baseline, illustrating the contrast with SpeechGR' s transcript-free, index-free approach.

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Category	Model	Training	Retrieval Setup & Description
Sparse text	BM25	None	TF-IDF lexical matching on ASR transcripts; classical bag-of-words baseline.
Speech dense	SpeechDPR (Lin et al., 2024)	Fine-tuned	HuBERT encoder projected to dense vectors with ANN index; state-of-the-art speech-only retriever.
Index-free speech	SpeechGR (this work)	Fine-tuned	End-to-end generative model using discrete units, no transcripts or indices.

Table 4.1: Baseline models evaluated against SpeechGR.

4.3 Evaluation

Evaluation aligns with the SpeechDPR framework (Lin et al., 2024), where each SLUE-SQA-5 spoken query is paired with a single correct passage. Performance is assessed using Top-1 accuracy (Hit@1) and Top-20 accuracy (Hit@20), defined as the proportion of queries where the gold passage identifier ranks within the top 1 or 20 predictions. Headline comparisons in Section 4.6 are conducted on the test split, while ablation studies in Section 4.5 use the development split initially, advancing only configurations showing promise to the test set.

4.4 Model Configuration

The primary SpeechGR model employs Flan-T5 base (Chung et al., 2024) as its backbone, chosen for optimal performance after initial tests. The vocabulary is expanded with one embedding per discrete unit, initialized from an existing text token's vector, with discrete

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units trained on the LibriSpeech 100h subset using features from HuBERT layer 22. Pre-training is conducted on a 6k-hour LibriLight (Kahn et al., 2020) subset for 10 epochs with a 15% span-masking ratio, using sentinel tokens for span reconstruction, and is tailored to the layer 22 features. Discrete units are extracted from HuBERT-large (Hsu et al., 2021) layer 22 and quantized into a 500-token codebook. The model is fine-tuned end-to-end on SLUE-SQA-5 with a cross-entropy loss, and inference uses constrained beam search with 20 beams. Ablation studies in Section 4.5 explore variations in backbone sizes, codebook parameters, Window-level Q-Former settings, and ranking-loss weights.

4.5 Main Results

Table 4.2 presents the primary retrieval results on the SLUE-SQA-5 test split, showing SpeechGR achieving 5.21% Hit@1 and 18.452% Hit@20 with Flan-T5 base, pre-trained for 10 epochs, and a 500-token codebook, closely approaching SpeechDPR's 19.73% Hit@20. BM25, using ground-truth text, retrieves almost no target passages. SpeechGR's transcript-free, index-free approach thus nears state-of-the-art speech-only retrieval, leveraging discrete units trained on LibriSpeech 100h with HuBERT layer 22 features.

The slight gap with SpeechDPR, which benefits from knowledge distillation boosting its Hit@20 to 19.73% (versus 0.04% without), highlights SpeechGR's potential despite its speech-only setup. The 10-epoch pre-training on 6k-hour LibriLight may be insufficient for full adaptation to the discrete-unit vocabulary, as a 0.7% Hit@1 improvement was observed with 69,000-step pre-training. Experiments with a 128-token limit did not enhance performance, suggesting that input length reduction may not address current limitations. Extending pre-training or scaling the dataset could further improve results, validating the

doi:10.6342/NTU202504440

feasibility of a generative, textless retrieval system.

Model	**Hit@1 (%)**	**Hit@20 (%)**
BM25	0.0	0.0
ASR + BM25	1.5	5.2
SpeechDPR [speechdpr]	-	19.73
SpeechDPR w/o KD	-	0.04
SpeechGR (this work)	5.81	18.452

Table 4.2: Main retrieval results on SLUE-SQA-5 test split for specified baselines and SpeechGR.

4.6 Ablation Studies

We conduct ablation studies to measure how each design choice affects retrieval performance. In every experiment, a single factor is changed while the rest of the system stays identical to the main configuration in § 4.4, with some of the experiments excluding speech-aware pre-training where noted. The individual results and their interpretations are reported below.

4.6.1 Effect of Discrete Unit Granularity

To examine how discrete unit granularity influences retrieval, we vary three factors:

(i) the Hubert layer that extracts the features,

(ii) the size of the k-means codebook, and

(iii) optional BPE segmentation on the larger vocabularies.

Following the setup of Wang et al. (2024), we explored two representation layers from HuBERT-large. Layer 7 primarily conveys phonetic detail, whereas layer 22 is known to encode higher-level semantics. For each layer, we trained k-means codebooks on the 960-

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hour LibriSpeech corpus (Kahn et al., 2020) with the configurations listed in Table 4.3. Some experiments were conducted without speech-aware pre-training to isolate granularity effects.

Our results differ from those reported by Wang et al. The best development accuracy is obtained with $K = 500$ at layer 22; the $K = 2000$, which they found optimal, yields lower performance in our retrieval task. Across all K values, layer 22 consistently outperforms layer 7, confirming that deeper HuBERT activations are better suited to semantic retrieval. BPE segmentation generally reduced performance, suggesting that subword merges remove useful information. Based on these findings, all subsequent experiments use the layer 22, $K = 500$ configuration.

Layer	**K**	**BPE Vocab Size**	**Hit@1 (%)**	**Hit@20 (%)**
7	500	None	1.2	6
22	128	None	3.76	13.04
22	500	None	**3.61**	**14.189**
22	1000	None	3.44	12.72
22	500	1000	1.85	9.78
22	1000	2000	1.97	9.11
22	2000	6000	1.47	7.68

Table 4.3: Effect of discrete unit granularity on retrieval performance.

4.6.2 Identifier Design

To evaluate the impact of identifier design on retrieval performance, we conducted an ablation study comparing two approaches: string-based identifiers and atomic identifiers. String-based identifiers represent documents as multi-token sequences, preserving contextual and structural information from the dataset, similar to approaches in prior generative retrieval frameworks (Tay et al., 2022). Atomic identifiers, conversely, assign each document a unique single-token code derived from a learned embedding space, aiming for a

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compact representation. This approach was motivated by recent work in cross-modal generative retrieval, such as GRACE (Li et al., 2024b), which achieved superior performance using atomic identifiers for non-text (image) passages, demonstrating their effectiveness in capturing semantic relevance for multimodal inputs.

We evaluated both identifier designs on the SLUE-SQA-5 dataset using the default SpeechGR configuration. String-based identifiers achieved a Hit@1 of 3.61% and Hit@20 of 14.19%, while atomic identifiers yielded a Hit@1 of 4.54% and Hit@20 of 13.69%. Although atomic identifiers outperformed string-based identifiers in top-1 accuracy, indicating better precision for the most relevant document, string-based identifiers excelled in top-20 accuracy, suggesting improved recall for a broader set of relevant documents. In contrast to GRACE, where atomic identifiers were optimal for image-based retrieval due to their compact representation, our results indicate that the multi-token nature of string-based identifiers better captures the complex acoustic and contextual patterns in discrete acoustic tokens for speech-based retrieval. Consequently, string-based identifiers were adopted in the final SpeechGR configuration to prioritize broader retrieval accuracy, which is critical for effective speech-to-speech retrieval in diverse linguistic contexts.

4.6.3 Impact of Speech-Aware Pre-training

We examine the benefit of additional speech-aware pre-training by comparing three configurations that differ only in the number of pre-training epochs on the 6 k-hour LibriLight subset with span corruption mask ratio to 15%. Zero epochs corresponds to directly fine-tuning on SLUE-SQA-5 for the GR task, while three and ten epochs add progressively more unsupervised exposure. Moving from 0 to 3 epochs gives about a 0.93-point gain in Hit@1 and roughly a 1.54-point gain in Hit@20. Extending to ten epochs provides a

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further increase of 1.27 points in Hit@1 and 2.72 points in Hit@20. These results suggest that speech-aware pre-training is helpful for further fine-tuning, and we retain the ten-epoch setting for subsequent experiments.

Pre-train epochs	**Hit@1 (%)**	**Hit@20 (%)**
0	3.61	14.189
3	4.54	15.73
10	**5.81**	**18.452**

Table 4.4: Effect of speech-aware pre-training epoch.

4.6.4 Ranking Loss

Motivated by recent work showing that an explicit ranking loss can sharpen generative-retrieval models—most notably RIPOR (Zeng et al., 2023) and LTRGR (Sun et al., 2023)—we assess its impact by adding a margin-based pairwise loss. We use the PyTorch implementation (`torch.nn.MarginRankingLoss`) with an in-batch negative strategy: for each spoken query, the gold identifier $\mathbf{y}^+$ is paired with every other identifier $\mathbf{y}^-$ in the same mini-batch, yielding an in-batch set of negatives. A candidate identifier $\mathbf{y}$ is assigned the similarity

$$S(\mathbf{q}, \mathbf{y}) = \sum_{t=1}^{L} (h_t \cdot e_{y_t}),$$

where h_t is the decoder hidden state at position t and e_{y_t} is the corresponding output-token embedding. The margin ranking loss for a single positive-negative pair is

$$\mathcal{L}_{rank} = \max\{0, m - [S(\mathbf{q}, \mathbf{y}^+) - S(\mathbf{q}, \mathbf{y}^-)]\},$$

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with hyperparameter m. When used, the overall objective becomes

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{rank},$$

where λ sets the weight of the ranking component. Using the default configuration (layer 22, $K = 500$), we compare models with and without the ranking loss. The version without ranking loss achieves Hit@1 of 3.61% and Hit@20 of 14.189%, while adding the ranking loss reduces performance, lowering Hit@1 by approximately 0.88% and Hit@20 by about 1.349%. Although the ranking loss accelerates convergence, it hurts final retrieval accuracy, so we discard it in all other experiments.

4.6.5 Window-level Q-Former

To investigate whether compressing input sequences improves retrieval performance, we conducted an ablation study by adding a window-level Q-Former (Query Transformer) module before the T5 encoder, inspired by SpeechLLM configurations such as COSMIC and SALMONN (Pan et al., 2023; Tang et al., 2023a). The Q-Former, a transformer-based module, processes discrete acoustic tokens from HuBERT (Section 3.1) into a shorter sequence of query embeddings, aiming to capture essential acoustic features while reducing computational complexity.

Given a sequence of discrete acoustic tokens $\mathbf{X} = [x_1, x_2, \ldots, x_T]$ derived from HuBERT features and k-means clustering, the Q-Former first converts $\mathbf{X}$ into frame-level embeddings via a lookup in the discrete-unit embedding matrix:

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$$\mathbf{M} = Embed(\mathbf{X}) \in \mathbb{R}^{T\times d}.$$

The embedding sequence $\mathbf{M}$ is then split into non-overlapping chunks $\mathbf{M}_\ell \in \mathbb{R}^{L\times d}$, where $L = 17$ is the window length. For each chunk $\mathbf{M}_\ell$, a set of $Q = 1$ query vectors $\mathbf{Q} \in \mathbb{R}^{Q\times d}$ attends to the discrete units via cross-attention:

$$\mathbf{Z}_\ell = QFormer(\mathbf{Q}, \mathbf{M}_\ell) \in \mathbb{R}^{Q\times d},$$

producing one embedding vector per window. Concatenating the embeddings from all windows yields a compressed sequence:

$$\mathbf{Z} = [\mathbf{Z}_1, \ldots, \mathbf{Z}_{T/L}] \in \mathbb{R}^{\frac{T}{L}\times d}.$$

This shorter sequence $\mathbf{Z}$ replaces the original frame-level embeddings $\mathbf{M}$ as input to the T5 encoder for identifier generation. By segmenting the input into windows and applying cross-attention, the Q-Former reduces the sequence length by a factor of L, potentially lowering computational demands while preserving local acoustic patterns.

In experiments on the SLUE-SQA-5 dataset, we compared the baseline SpeechGR model (without Q-Former) to the version with the Q-Former, using a window length of $L = 17$ and $Q = 1$, following Pan et al. (2023). The Q-Former configuration resulted in a performance drop, with Hit@1 decreasing from 3.61% to 2.73% and Hit@20 from 14.189% to 12.584%. Although the Q-Former compresses the input sequence, the loss of fine-grained acoustic information likely hinders precise identifier generation. Due to this performance

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degradation and the added training complexity, the Q-Former was not included in the final SpeechGR configuration, making this an exploratory ablation study.

4.6.6 Voice-Activity Detection (VAD)

Some query recordings in SLUE-SQA-5 are empty or filled with long pauses. To test whether removing non-speech regions helps retrieval, we re-processed every query and passage with the Silero VAD toolkit (2024), cropping segments whose probability of speech fell below the default threshold. After re-training under otherwise default setup, performance dropped slightly: Hit@1 fell from 3.61 % to 3.27 %, and Hit@20 from 14.189 % to 12.97 %.

A likely explanation is that silent intervals already collapse into a single, low-frequency discrete token that is removed by our deduplication step; eliminating the silences therefore provides little new signal while shortening some queries unevenly. Because VAD yields no benefit and incurs extra preprocessing, we keep the raw audio in the final SpeechGR pipeline.

4.7 Summary

SpeechGR, evaluated on the SLUE-SQA-5 test split, achieves 5.81% Hit@1 and 18.452% Hit@20 using FlanT5-base, 10-epoch pre-training on 6k-hour LibriLight, and a 500-token codebook, nearing SpeechDPR's 19.73% Hit@20. Ablation studies show HuBERT layer 22 with $K = 500$ outperforms layer 7 and BPE variants, while 10-epoch pre-training boosts Hit@1 by 2.2 points and Hit@20 by 4.263 points. Ranking loss, Q-Former, and VAD reduce accuracy and are excluded. These results confirm SpeechGR' s effectiveness

doi:10.6342/NTU202504440

as a textless, index-free retrieval system, with potential for improvement via extended pre-training or larger datasets.

doi:10.6342/NTU202504440

doi:10.6342/NTU202504440

Chapter 5 Conclusion & Future Work

5.1 Conclusion

This work presents **SpeechGR**, a pioneering end-to-end generative retrieval system that processes spoken queries and passages directly, bypassing text transcripts and external indices. Motivated by the scarcity of textual resources for over half of the world's 7,000+ languages (2025), SpeechGR leverages discrete acoustic tokens and a generative language model to produce document identifiers, addressing limitations in traditional retrieval systems. Evaluated on the SLUE-SQA-5 dataset, SpeechGR achieves a Hit@1 of 5.81% and Hit@20 of 18.452% with FlanT5-base, 10-epoch pre-training, and a 500-token codebook, closely approaching SpeechDPR's 19.73% Hit@20 (Lin et al., 2024). This performance, validated by ablation studies showing the efficacy of layer 22 features and pre-training, underscores its potential as a textless, speech-only solution.

The contributions are threefold: (1) a viable framework for end-to-end speech generative retrieval, eliminating ASR dependency; (2) advancement in textless NLP for unwritten languages; and (3) insights into optimizing generative retrieval, notably the benefits of deeper layers and pre-training. Despite these advances, limitations persist: the English-focused SLUE-SQA-5 dataset restricts generalizability, and the computational demands of large models pose challenges for low-resource settings.

doi:10.6342/NTU202504440

5.2 Future Work

SpeechGR opens several research directions. First, scaling to diverse languages, especially unwritten ones, requires collecting datasets like Common Voice (Ardila et al., 2020) and adapting discrete unit extraction to varied phonologies. Second, improving scale pre-training with larger audio datasets could enhance the language model's understanding of acoustic patterns, potentially boosting retrieval performance. Third, exploring the feasibility of decoder-only generative retrieval (GR) could facilitate integration with SpeechLLM, leveraging its architecture for more efficient and context-aware speech processing. These advancements would support the development of inclusive, supervision-free retrieval systems for global access.

doi:10.6342/NTU202504440

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